

TuniCars+

Resilience Testing Report

Docker Swarm vs Kubernetes

Session ID

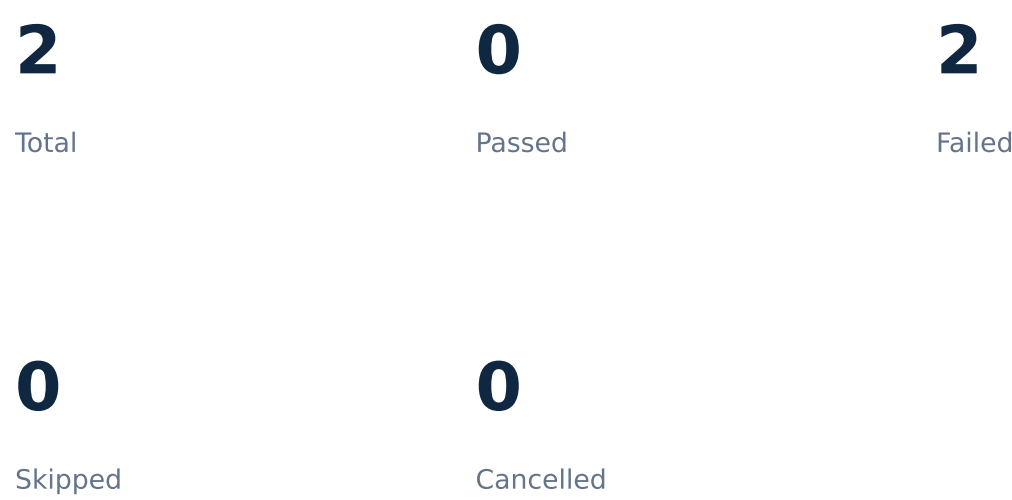
20260813_215453_c2f034

Date

2026-08-13T20:54:53.834259+00:00

Executive Summary

20260813_215453_c2f034



Overall measured availability: 99.37%

During this campaign, 0 of 2 scenarios passed, 2 failed and 0 were skipped. Conclusions are limited to the measured scenarios and recorded environment constraints.

Test Environment

Architecture and technologies

Frontend → Backend → PostgreSQL

Docker Swarm environment

- Docker services
- FastAPI backend
- Next.js frontend
- PostgreSQL database

Kubernetes / Minikube environment

- Kubernetes Deployments and Services
- Minikube nodes
- FastAPI · Next.js · PostgreSQL

Test Scenarios

Scenario	swarm
memory	2

Full Results Table

Scenario	Platform	Detection	Recovery	Required	Success	Avg ms	P95 ms	HTTP Fail	Avail. %	Status	Skip Reason	Error Message
memory	swarm	N/A	N/A	N/A	N/A	38806.074	381.94	1	98.73	FAIL	N/A	Swarm stress container exceeded its scenario deadline.
memory	swarm	N/A	N/A	N/A	N/A	12.928	21.619	N/A	100.0	FAIL	N/A	Swarm stress container exceeded its scenario deadline.

swarm — memory

State: Fail
Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: 1

Availability (%): 98.73

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

Swarm stress container exceeded its scenario deadline.

swarm — memory

State: Fail
Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): 100.0

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

Swarm stress container exceeded its scenario deadline.

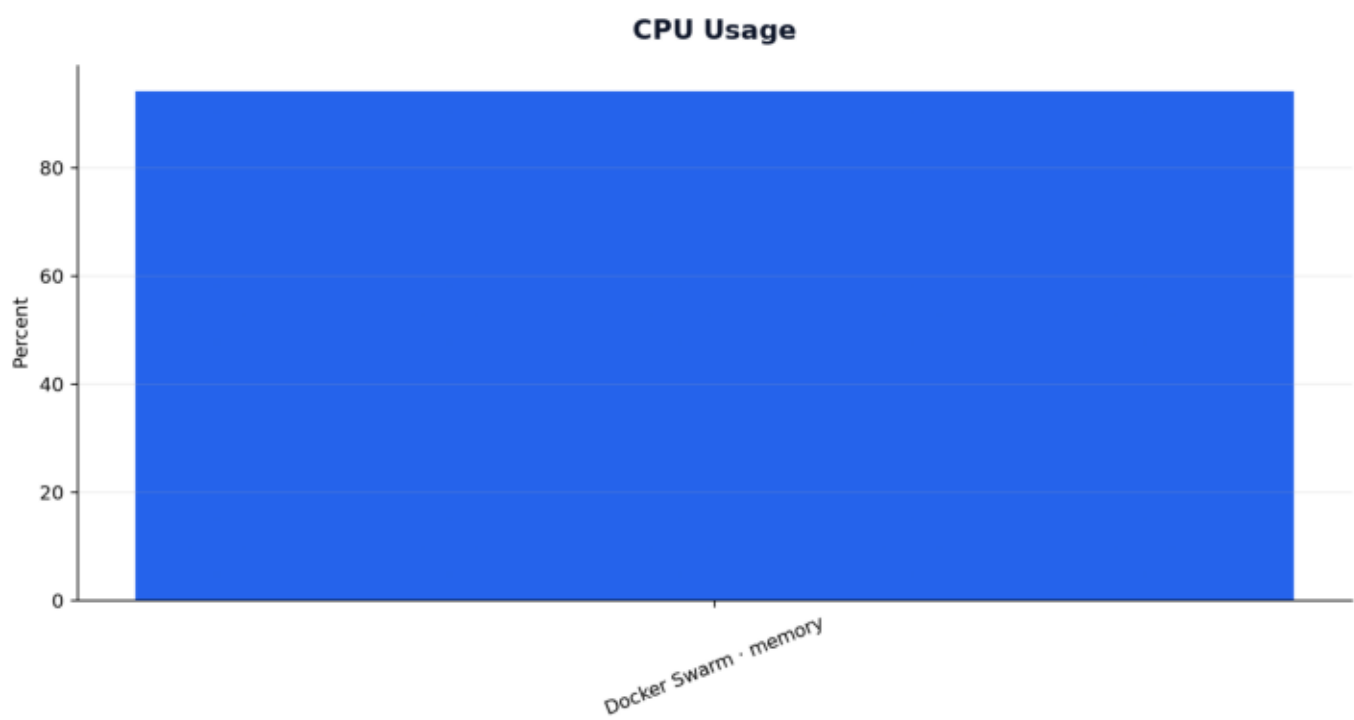
Availability

Current-session measured data



Cpu Usage

Current-session measured data



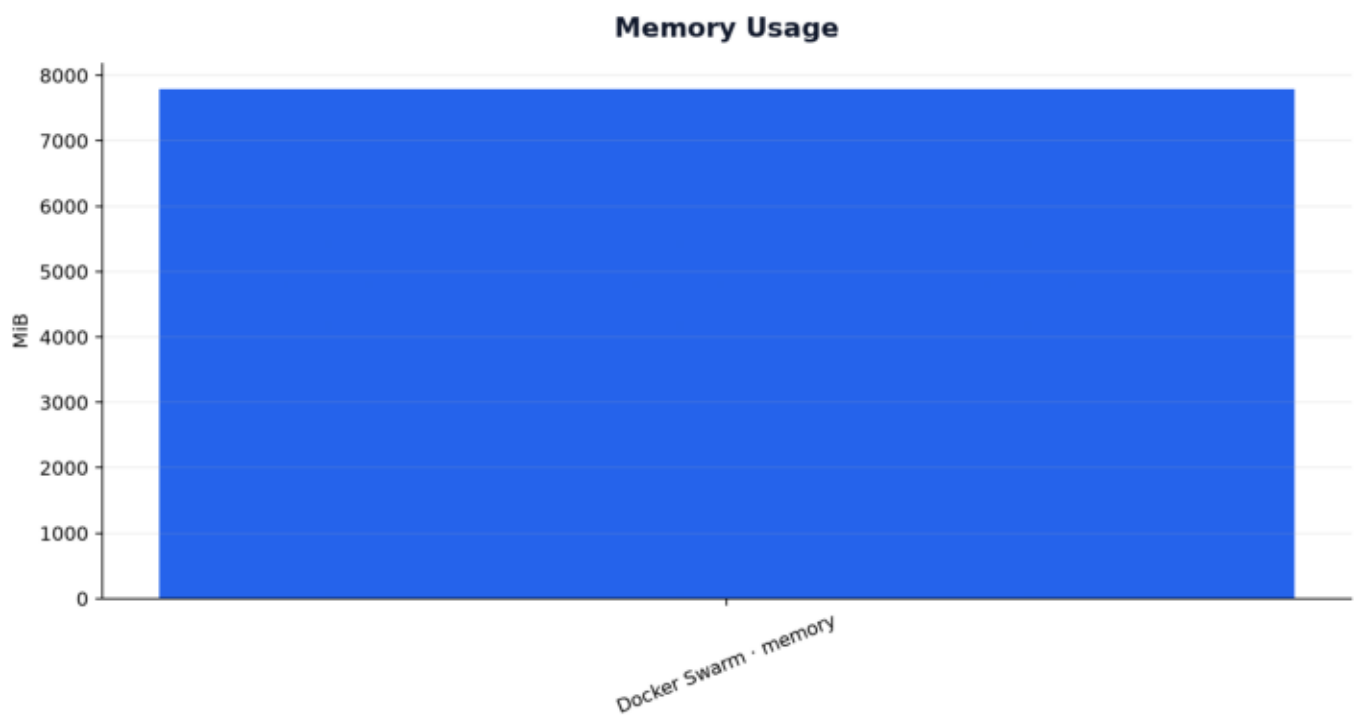
Http Failures

Current-session measured data



Memory Usage

Current-session measured data



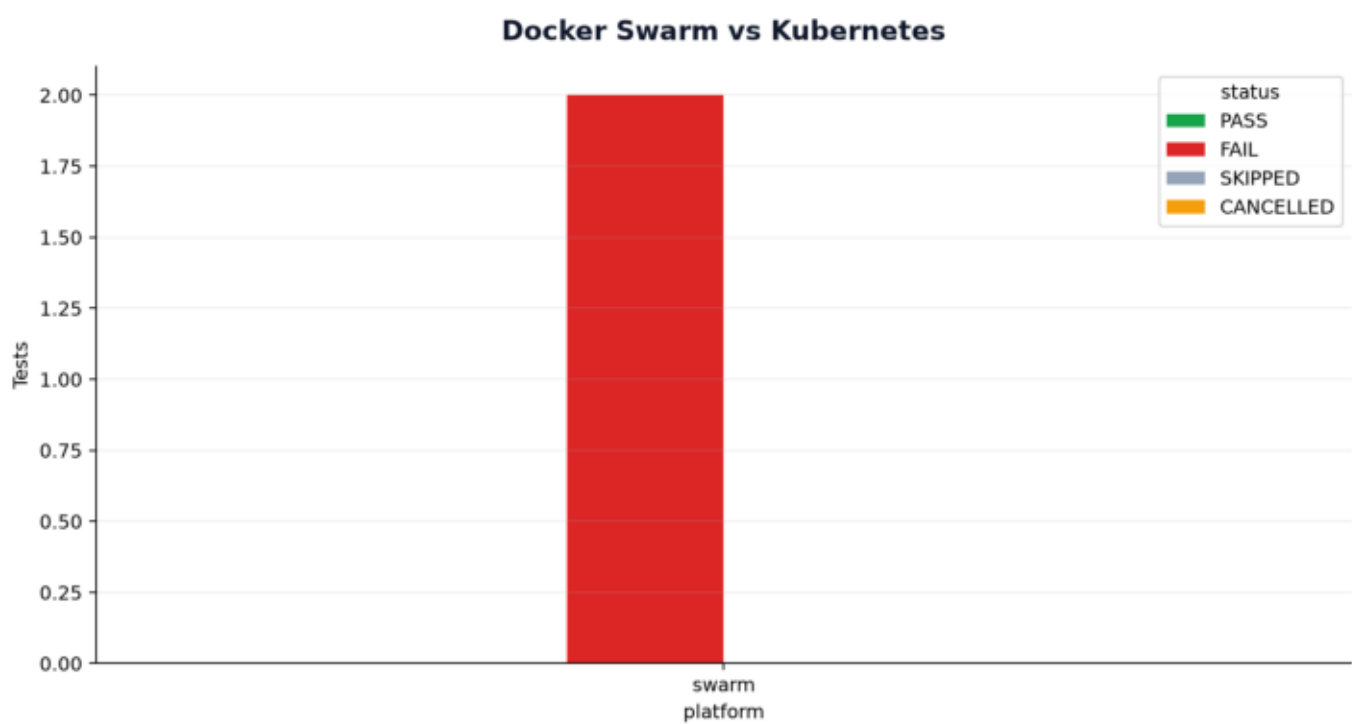
P95 Response Time

Current-session measured data



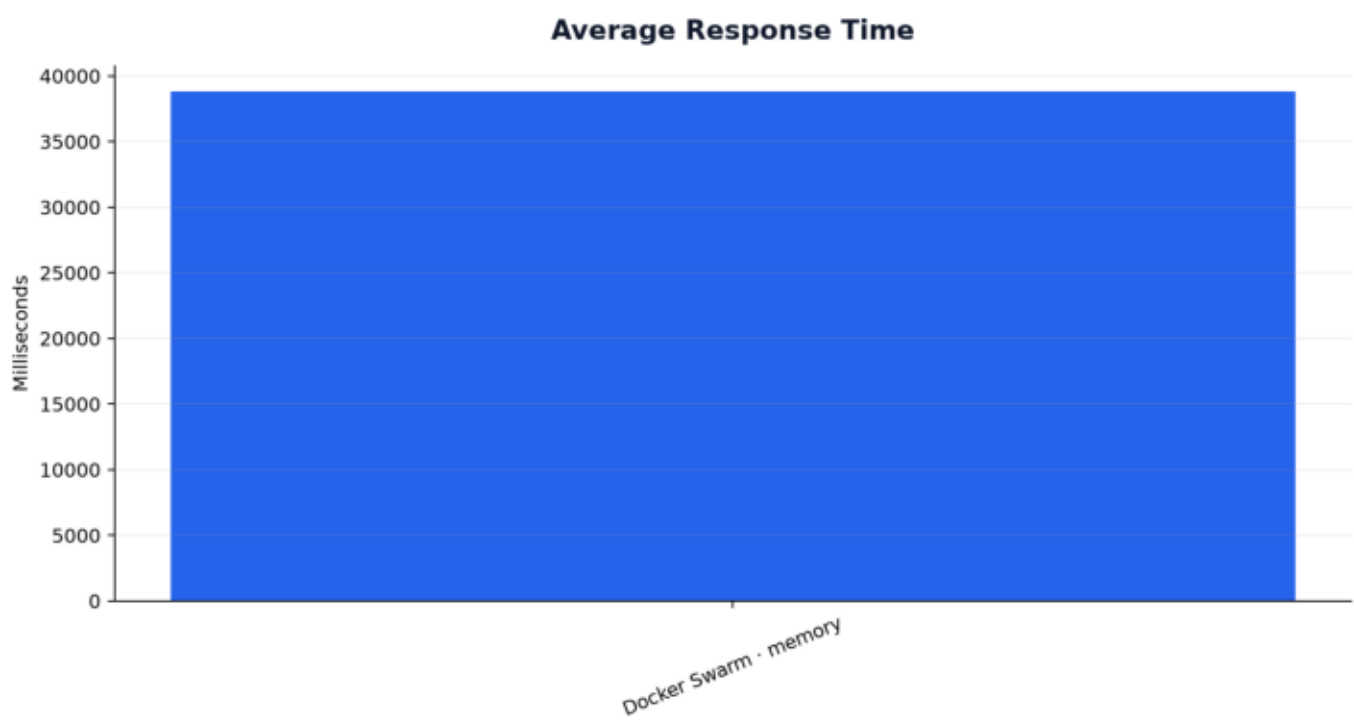
Platform Comparison

Current-session measured data



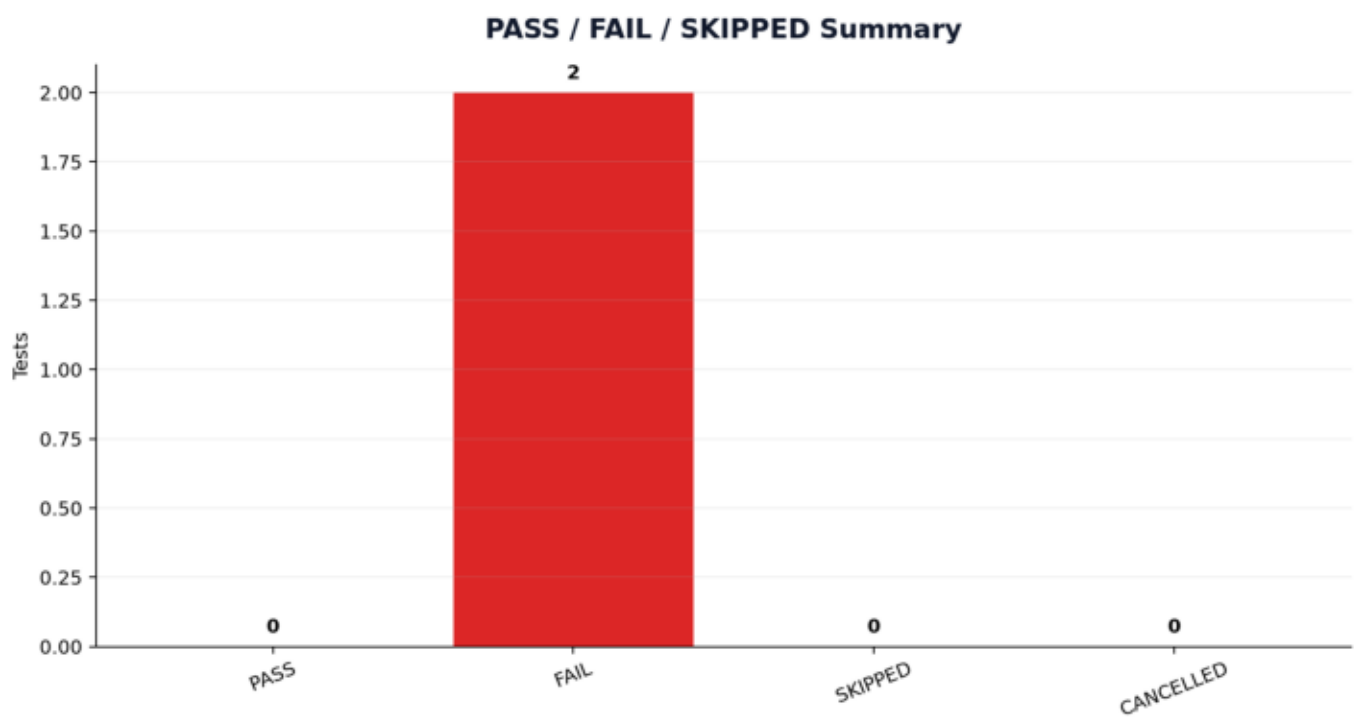
Response Time

Current-session measured data



Status Summary

Current-session measured data



Failures and Errors

Platform	Scenario	Status	Error
swarm	memory	FAIL	Swarm stress container exceeded its deadline.
swarm	memory	FAIL	Swarm stress container exceeded its deadline.

Skipped Tests

Platform	Scenario	Exact reason
—	—	No skipped tests recorded.

Swarm vs Kubernetes Comparison

Platform	FAIL
swarm	2

Final Conclusion

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During this campaign, 0 of 2 scenarios passed, 2 failed and 0 were skipped. Conclusions are limited to the measured scenarios and recorded environment constraints.